NETWORK SECURITY

Subject: Cryptography and Network Security

Assignment

Q1. Define network security and explain its fundamental principles and objectives. Discuss the importance of confidentiality, integrity, availability, authentication, and authorization in securing network systems.

Network Security means protecting computer networks and data from unauthorized access, attacks and damage.

Its main objective is to keep information safe.

Fundamental Principles

- **Confidentiality:** Ensures that only authorized people can access information.
- **Integrity:** Ensures that data is not changed without permission.
- **Availability:** Ensures that network services are available when needed.
- **Authentication:** Verifies the identity of a user.
- **Authorization:** Gives users permission to access specific resources.

Example: Online banking uses passwords and OTPs to protect customer information and transactions.

Q2. Explain the threat landscape in network security. Discuss different types of network attacks and vulnerabilities, along with suitable examples and preventive measures.

The threat landscape refers to different attacks and risks that can damage computer networks and steal information.

Types of Network Attacks

- **Malware:** Harmful software that damages systems.
- **Phishing:** Fake emails or websites used to steal passwords.
- **DoS Attack:** Makes a network service unavailable.
- **Man-in-the-Middle Attack:** Secretly intercepts communication.
- **Password Attack:** Attempts to steal or guess passwords.

Preventive Measures

- Use strong passwords.
- Install antivirus software.
- Use firewalls.
- Keep software updated.
- Avoid suspicious links.

Example: A fake banking email may steal a user's login details.

Q3. Discuss the legal and ethical considerations in network security. Explain the importance of data privacy, responsible security testing, authorized access, and compliance with applicable cybersecurity laws.

Legal and ethical considerations mean following laws and behaving responsibly while protecting computer networks and information.

Important Considerations

- **Data Privacy:** Personal information should be protected from unauthorized access.
- **Responsible Testing:** Security testing should be performed carefully without damaging systems.
- **Authorized Access:** Users should access only those systems for which they have permission.
- **Legal Compliance:** Organizations must follow applicable cybersecurity and data protection laws.
- **Ethical Behaviour:** Security professionals should not misuse confidential information.

Example: A security tester must take permission before testing a company's website.

Conclusion: Legal and ethical practices help protect users and build trust.

Q4. Explain the role of cryptography in securing network communications. Describe the working and security features of SSL/TLS and IPsec protocols.

Cryptography is the process of protecting information by converting readable data into a secure form. It helps protect communication over networks.

1. SSL/TLS

- Used to secure communication between browsers and websites.
- Encrypts transmitted information.
- Uses digital certificates to verify website identity.
- TLS is the modern and secure replacement for SSL.

Example: HTTPS websites use TLS to protect user information.

2. IPsec

- Provides security at the IP network layer.
- Encrypts and authenticates IP packets.
- Protects data during transmission.
- Commonly used in VPNs.

Conclusion: Both protocols help provide secure network communication.

Q5. What is Public Key Infrastructure (PKI)? Explain its components and working. Discuss the role of digital certificates and Certificate Authorities in establishing trust and securing network communications.

Public Key Infrastructure (PKI) is a security system that uses public keys, private keys and digital certificates to establish trust and secure communication.

Main Components

- **Public Key:** Used to verify signatures or perform encryption.
- **Private Key:** Kept secret by its owner.
- **Digital Certificate:** Connects a public key with an identity.
- **Certificate Authority (CA):** Issues and verifies digital certificates.
- **Registration Authority (RA):** Helps verify the identity of certificate applicants.

Working

- The user requests a digital certificate.
- The CA verifies the identity.
- The CA issues the certificate.
- Other users verify the certificate to establish trust.

Example: HTTPS websites use digital certificates to prove their identity.